

1D CNN — Architecture Variant Comparison

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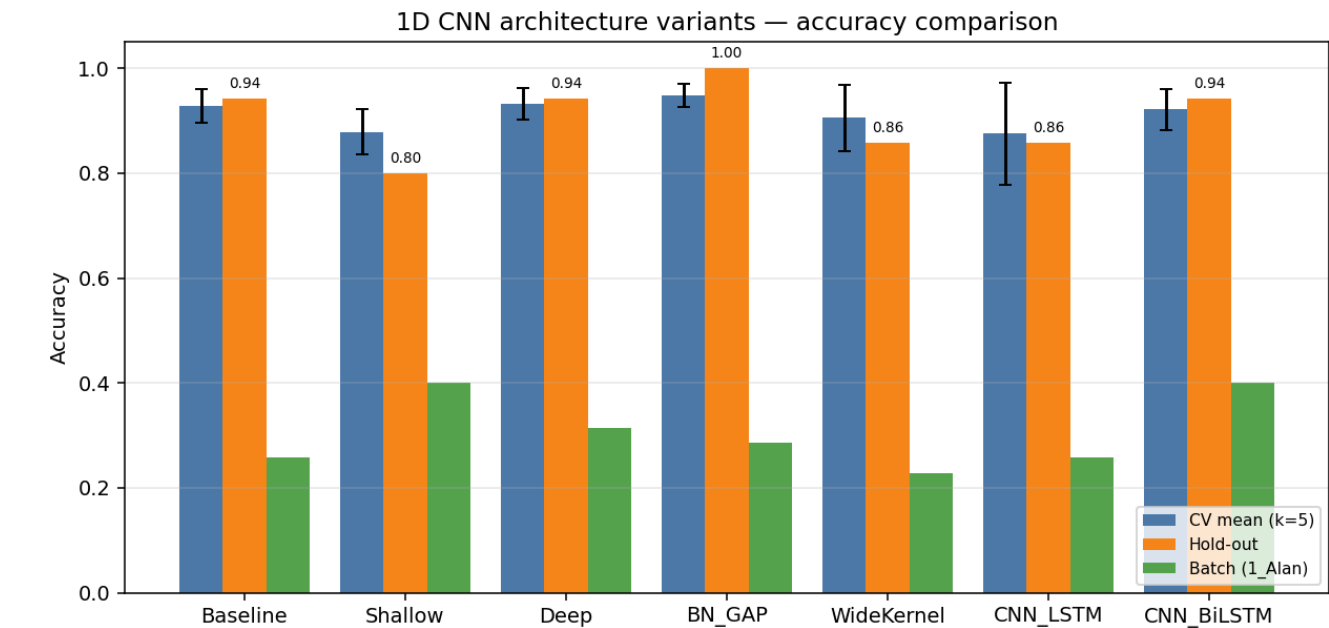
Data & configuration

Training path 1	/home/jestin/ThesisRepo/ML/NewTestData/2_Alex/Dynamic
Training path 2	/home/jestin/ThesisRepo/ML/NewTestData/3_Anghad/Dynamic
Training path 3	/home/jestin/ThesisRepo/ML/NewTestData/4_Daniel/Dynamic
Training path 4	/home/jestin/ThesisRepo/ML/NewTestData/6_Harry/Dynamic
Training path 5	/home/jestin/ThesisRepo/ML/NewTestData/8_Jestin/Dynamic
Training path 6	/home/jestin/ThesisRepo/ML/NewTestData/11_Mansh/Dynamic
Training path 7	/home/jestin/ThesisRepo/ML/NewTestData/12_Marcus/Dynamic
Training path 8	/home/jestin/ThesisRepo/ML/NewTestData/14_StephenV2/Dynamic
Training path 9	/home/jestin/ThesisRepo/ML/NewTestData/15_Tash/Dynamic
Training path 10	/home/jestin/ThesisRepo/ML/NewTestData/17_Raquel/Dynamic
Batch-test root	/home/jestin/ThesisRepo/ML/NewTestData/1_Alan/Dynamic
Total trials	341 (train pool 306, hold-out 35)
Classes	7: Double_Lower, Double_Nothing, Double_Pistol_Recoil, Double_Raise, Double_Wiggle, Double_cmere, Drum_Roll
Sensor channels	176 (sequence length 90)
Resample / filter	90 steps · Butterworth cutoff 15.0 Hz, order 4
Normalisation	minmax
CV folds · shuffle	5 · True
Augmentation	on · copies=5 · amplitude_scale, baseline_offset
Epochs · batch size	15 · 16

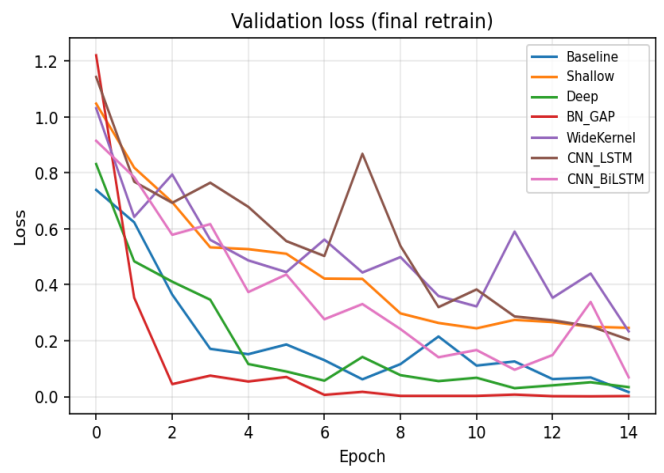
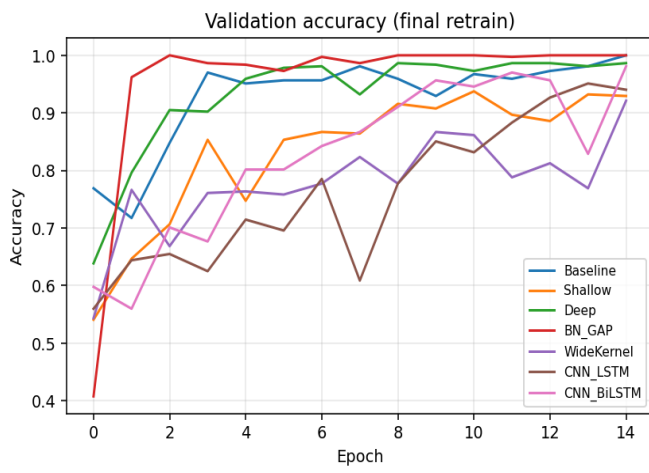
Variant comparison (summary)

Variant	Params	CV mean (k=5)	CV std	Hold-out acc	Hold-out loss	Batch acc	Batch n
Baseline	113,255	0.9281	0.0320	0.9429	0.0879	0.2571	9/35
Shallow	72,487	0.8790	0.0425	0.8000	0.3679	0.4000	14/35
Deep	204,007	0.9313	0.0302	0.9429	0.1156	0.3143	11/35
BN_GAP	57,319	0.9477	0.0218	1.0000	0.0007	0.2857	10/35
WideKernel	131,687	0.9052	0.0635	0.8571	0.3566	0.2286	8/35
CNN_LSTM	68,455	0.8756	0.0972	0.8571	0.4173	0.2571	9/35
CNN_BiLSTM	105,575	0.9216	0.0393	0.9429	0.1732	0.4000	14/35

Best on hold-out: **BN_GAP** · Best on CV: **BN_GAP**



Training curves (final retrain on full training pool)



Variant: Baseline

Conv1D(32,k=4)→Pool→Conv1D(64,k=4)→Pool→Flatten→Dense(64)→Dropout(0.3)→Softmax

Trainable params	113,255
CV mean ± std	0.9281 ± 0.0320
Hold-out test acc / loss	0.9429 · 0.0879
Batch-test acc	0.2571 (9/35)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1464	62	0.9194	0.9677	0.9194
2	1470	61	0.9180	0.9344	0.9180
3	1470	61	0.9836	1.0000	0.9836
4	1470	61	0.8852	0.9344	0.8852
5	1470	61	0.9344	0.9508	0.9344

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	5
Double_Nothing	1.000	0.600	0.750	5
Double_Pistol_Recoil	1.000	1.000	1.000	5
Double_Raise	1.000	1.000	1.000	5
Double_Wiggle	0.714	1.000	0.833	5
Double_cmere	1.000	1.000	1.000	5
Drum_Roll	1.000	1.000	1.000	5
macro avg	0.959	0.943	0.940	35
weighted avg	0.959	0.943	0.940	35

Batch test per class (1_Alan)

Class	Correct	Total	Accuracy
Double_Lower	1	5	0.200
Double_Nothing	3	5	0.600
Double_Pistol_Recoil	0	5	0.000
Double_Raise	5	5	1.000
Double_Wiggle	0	5	0.000
Double_cmere	0	5	0.000
Drum_Roll	0	5	0.000
Overall	9	35	0.257

Variant: Shallow

Single Conv block: Conv1D(32,k=5)→Pool→Flatten→Dense(32)→Dropout(0.3)→Softmax

Trainable params	72,487
CV mean ± std	0.8790 ± 0.0425
Hold-out test acc / loss	0.8000 · 0.3679
Batch-test acc	0.4000 (14/35)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1464	62	0.9194	0.9839	0.9194
2	1470	61	0.9180	0.9180	0.9180
3	1470	61	0.9016	0.9344	0.9016
4	1470	61	0.8197	0.8525	0.8197
5	1470	61	0.8361	0.9016	0.8361

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	0.800	0.889	5
Double_Nothing	0.500	0.600	0.545	5
Double_Pistol_Recoil	1.000	0.600	0.750	5
Double_Raise	1.000	0.800	0.889	5
Double_Wiggle	0.556	1.000	0.714	5
Double_cmere	1.000	1.000	1.000	5
Drum_Roll	1.000	0.800	0.889	5
macro avg	0.865	0.800	0.811	35
weighted avg	0.865	0.800	0.811	35

Batch test per class (1_Alan)

Class	Correct	Total	Accuracy
Double_Lower	0	5	0.000
Double_Nothing	4	5	0.800
Double_Pistol_Recoil	0	5	0.000
Double_Raise	5	5	1.000
Double_Wiggle	0	5	0.000
Double_cmere	5	5	1.000
Drum_Roll	0	5	0.000
Overall	14	35	0.400

Variant: Deep

Three Conv blocks: Conv1D(32,4)→Conv1D(64,4)→Conv1D(128,3)→Flatten→Dense(128)→Dropout(0.4)→Softmax

Trainable params	204,007
CV mean ± std	0.9313 ± 0.0302
Hold-out test acc / loss	0.9429 · 0.1156
Batch-test acc	0.3143 (11/35)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1464	62	0.9677	1.0000	0.9677
2	1470	61	0.9016	0.9344	0.9016
3	1470	61	0.9180	1.0000	0.9180
4	1470	61	0.9016	0.9180	0.9016
5	1470	61	0.9672	0.9672	0.9672

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	5
Double_Nothing	1.000	0.600	0.750	5
Double_Pistol_Recoil	1.000	1.000	1.000	5
Double_Raise	1.000	1.000	1.000	5
Double_Wiggle	0.714	1.000	0.833	5
Double_cmere	1.000	1.000	1.000	5
Drum_Roll	1.000	1.000	1.000	5
macro avg	0.959	0.943	0.940	35
weighted avg	0.959	0.943	0.940	35

Batch test per class (1_Alan)

Class	Correct	Total	Accuracy
Double_Lower	1	5	0.200
Double_Nothing	5	5	1.000
Double_Pistol_Recoil	0	5	0.000
Double_Raise	5	5	1.000
Double_Wiggle	0	5	0.000
Double_cmere	0	5	0.000
Drum_Roll	0	5	0.000
Overall	11	35	0.314

Variant: BN_GAP

Conv+BN+ReLU ×3 with GlobalAveragePooling1D head — fewer params, less overfit

Trainable params	57,319
CV mean ± std	0.9477 ± 0.0218
Hold-out test acc / loss	1.0000 · 0.0007
Batch-test acc	0.2857 (10/35)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1464	62	0.9516	0.9677	0.9516
2	1470	61	0.9344	0.9344	0.9344
3	1470	61	0.9836	1.0000	0.9836
4	1470	61	0.9180	0.9180	0.9180
5	1470	61	0.9508	0.9672	0.9508

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	5
Double_Nothing	1.000	1.000	1.000	5
Double_Pistol_Recoil	1.000	1.000	1.000	5
Double_Raise	1.000	1.000	1.000	5
Double_Wiggle	1.000	1.000	1.000	5
Double_cmere	1.000	1.000	1.000	5
Drum_Roll	1.000	1.000	1.000	5
macro avg	1.000	1.000	1.000	35
weighted avg	1.000	1.000	1.000	35

Batch test per class (1_Alan)

Class	Correct	Total	Accuracy
Double_Lower	0	5	0.000
Double_Nothing	4	5	0.800
Double_Pistol_Recoil	0	5	0.000
Double_Raise	5	5	1.000
Double_Wiggle	0	5	0.000
Double_cmere	1	5	0.200
Drum_Roll	0	5	0.000
Overall	10	35	0.286

Variant: WideKernel

Conv1D(32,k=8)→Pool→Conv1D(64,k=8)→Pool→Flatten→Dense(64)→Dropout(0.3)→Softmax

Trainable params	131,687
CV mean ± std	0.9052 ± 0.0635
Hold-out test acc / loss	0.8571 · 0.3566
Batch-test acc	0.2286 (8/35)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1464	62	0.9194	0.9194	0.9194
2	1470	61	0.9016	0.9180	0.9016
3	1470	61	0.9672	0.9836	0.9672
4	1470	61	0.7869	0.8525	0.7869
5	1470	61	0.9508	0.9508	0.9508

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	0.500	0.600	0.545	5
Double_Nothing	1.000	0.600	0.750	5
Double_Pistol_Recoil	1.000	1.000	1.000	5
Double_Raise	1.000	1.000	1.000	5
Double_Wiggle	0.667	0.800	0.727	5
Double_cmere	1.000	1.000	1.000	5
Drum_Roll	1.000	1.000	1.000	5
macro avg	0.881	0.857	0.860	35
weighted avg	0.881	0.857	0.860	35

Batch test per class (1_Alan)

Class	Correct	Total	Accuracy
Double_Lower	0	5	0.000
Double_Nothing	3	5	0.600
Double_Pistol_Recoil	0	5	0.000
Double_Raise	5	5	1.000
Double_Wiggle	0	5	0.000
Double_cmere	0	5	0.000
Drum_Roll	0	5	0.000
Overall	8	35	0.229

Variant: CNN_LSTM

Conv1D(32,4)→Pool→Conv1D(64,4)→Pool→Dropout→LSTM(64)→Dense(64)→Dropout→Softmax

Trainable params	68,455
CV mean ± std	0.8756 ± 0.0972
Hold-out test acc / loss	0.8571 · 0.4173
Batch-test acc	0.2571 (9/35)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1464	62	0.9516	0.9839	0.9516
2	1470	61	0.6885	0.8689	0.6885
3	1470	61	0.9508	1.0000	0.9508
4	1470	61	0.9016	0.9016	0.9016
5	1470	61	0.8852	0.9344	0.8852

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	5
Double_Nothing	0.600	0.600	0.600	5
Double_Pistol_Recoil	1.000	0.600	0.750	5
Double_Raise	1.000	1.000	1.000	5
Double_Wiggle	0.625	1.000	0.769	5
Double_cmere	1.000	1.000	1.000	5
Drum_Roll	1.000	0.800	0.889	5
macro avg	0.889	0.857	0.858	35
weighted avg	0.889	0.857	0.858	35

Batch test per class (1_Alan)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Nothing	2	5	0.400
Double_Pistol_Recoil	0	5	0.000
Double_Raise	1	5	0.200
Double_Wiggle	0	5	0.000
Double_cmere	1	5	0.200
Drum_Roll	0	5	0.000
Overall	9	35	0.257

Variant: CNN_BiLSTM

Conv1D(32,4)→Pool→Conv1D(64,4)→Pool→Dropout→BiLSTM(64)→Dense(64)→Dropout→Softmax

Trainable params	105,575
CV mean ± std	0.9216 ± 0.0393
Hold-out test acc / loss	0.9429 · 0.1732
Batch-test acc	0.4000 (14/35)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1464	62	0.9194	0.9839	0.9194
2	1470	61	0.9180	0.9180	0.9180
3	1470	61	0.9672	0.9672	0.9672
4	1470	61	0.8525	0.8525	0.8525
5	1470	61	0.9508	0.9508	0.9508

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	5
Double_Nothing	1.000	0.600	0.750	5
Double_Pistol_Recoil	1.000	1.000	1.000	5
Double_Raise	1.000	1.000	1.000	5
Double_Wiggle	0.714	1.000	0.833	5
Double_cmere	1.000	1.000	1.000	5
Drum_Roll	1.000	1.000	1.000	5
macro avg	0.959	0.943	0.940	35
weighted avg	0.959	0.943	0.940	35

Batch test per class (1_Alan)

Class	Correct	Total	Accuracy
Double_Lower	2	5	0.400
Double_Nothing	4	5	0.800
Double_Pistol_Recoil	1	5	0.200
Double_Raise	4	5	0.800
Double_Wiggle	0	5	0.000
Double_cmere	3	5	0.600
Drum_Roll	0	5	0.000
Overall	14	35	0.400